Assignment 6

CS 6117 (Fall 2025)

In this homework, we will consider how to solve more general recursive equations in categories. We will primarily be interested in categories of *domains*, which are special kinds of ordered sets. *Domain theory* studies many different kinds of domains, with the overall goal of giving denotational semantics for recursive programs (i.e., programs that may not terminate).

The primary reference for this homework is the classic paper by Smyth and Plotkin (SP), *The Category-Theoretic Solution of Recursive Domain Equations* (Siam JC, 1982). We recommend going through the paper if you are interested—and it should be possible to follow along given what we have covered in class—but you do *not* need to read the whole paper to do this homework. We will point out where to look as we go along. The survey *Fixed Points of Functors* (FPP) by Adámek, Milius, and Moss (especially Section 4) gives a cleaner presentation.

Exercise 1. Let's start by working with the category **Set**. An *ω-chain* is a countable set of objects $X_0, X_1, X_2 \cdots$ along with maps $f_i : X_i \to X_{i+1}$; it is a diagram from the category:

$$\omega \triangleq \bullet \to \bullet \to \cdots$$

Suppose we are trying to model the values of the untyped lambda calculus with two base values: true and false. A natural idea is to interpret this language in a set D such that:

$$D \cong \mathbb{B} \cup (D \to D)$$

where $\mathbb{B} = \{tt, ff\}$, $\cup$ is disjoint union, and $D \to D$ is the set of all functions from D to D.

1. Show that this equation does not have any solutions $D \in$ **Set**. [Hint: Cantor's theorem shows that for any set X, the set 2^X of all subsets of X must have strictly larger cardinality than X.]

2. (**Sparkly star**) Let's consider what happens if we try to solve the equation by iterating a functor and taking the colimit, like we did in class. Let $N :$ **Set** $\to$ **Set** be the functor defined by:

$$N(X) \triangleq \mathbb{N} \to X \quad \text{and} \quad N(f : X \to Y)(h : \mathbb{N} \to X) \triangleq f \circ h.$$

Show that N does not preserve colimits of ω-chains ("ω-colimits"). [Hint: consider the ω-chain $[0] \to [1] \to [2] \to \cdots$, where $[n] = \{0, 1, \ldots, n\}$ and the maps are inclusions.]

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1. Assume for contradiction that such a D exists. Let $|D| = \kappa$ be the cardinality of D. Then,$\kappa = |D| = |B| + |D \to D| = 2 + \kappa^\kappa$ If κ is finite, say $\kappa = n$, then $n = 2 + n^n$. For $n = 0$, $0 = 2 + 0 = 2$, false. For $n = 1$, $1 = 2 + 1 = 3$, false. For $n = 2$, $2 = 2 + 4 = 6$, false. For larger n, $n^n > n - 2$, so impossible. If κ is infinite, then $\kappa^\kappa = \max(\kappa, 2^\kappa) = 2^\kappa > \kappa$ by Cantor's theorem (the power set $2^\kappa > \kappa$). Thus,$\kappa = 2 + \kappa^\kappa = \kappa^\kappa > \kappa$, a contradiction. Therefore, no such D exists in **Set**.

2. Consider the functor $N : \mathbf{Set} \to \mathbf{Set}$, $N(X) = \mathbb{N} \to X$, $N(f)(h) = f \circ h$. The ω-chain is $[0] \to [1] \to \cdots$, with colimit $\mathbb{N}$ (union with inclusions).

   Applying N: $N([n]) = [n]^{\mathbb{N}}$, maps are post-compositions with inclusions, effectively inclusions of functions.

   The colimit of N's chain is $\bigcup_n [n]^{\mathbb{N}}$, the bounded functions $\mathbb{N} \to \mathbb{N}$ (image finite). But $N(\mathbb{N}) = \mathbb{N}^{\mathbb{N}}$, all functions, strictly larger. The induced map from colimit to $N(\mathbb{N})$ is inclusion, but not universal: e.g., compatible cocone to $\mathbb{N}$ sending bounded functions to their suprema doesn't factor uniquely through bounded functions, as unbounded are needed. Thus, N fails to preserve ω-colimits.

don't understand this sentence (could use more detail) but the idea is right

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Exercise 2. Now, we will move to a category of domains. This exercise is based on Example 2, Section 5 in SP, or Section 4 in FPP. A *complete partial order with bottom* (CPO) is a set with a partial order $(X, \sqsubseteq)$ that is complete: for any chain

$$x_0 \sqsubseteq x_1 \sqsubseteq \cdots$$

there is a least upper-bound of this chain in X. We write $\sqcup_{i \in \omega} x_i$ for the least upper-bound. Furthermore, each CPO has a least element $\bot \in X$, i.e., $\bot \sqsubseteq x$ for all $x \in X$.

The category $\mathbf{CPO}_\bot$ has objects CPOs with bottom and morphisms from $(X, \sqsubseteq_X) \to (Y, \sqsubseteq_Y)$ given by set functions $f : X \to Y$ that are:

1. Monotone: If $x \sqsubseteq_X x'$, then $f(x) \sqsubseteq_Y f(x')$
2. Complete: $\sqcup_{i \in \omega} f(x_i) = f(\sqcup_{i \in \omega} x_i)$
3. Strict: $f(\bot_X) = \bot_Y$.

Furthermore, the morphisms in $\mathbf{CPO}_\bot(X, Y)$ also carry a CPO structure: we can define an ordering on morphisms

$$f \sqsubseteq g \quad \text{iff} \quad f(x) \sqsubseteq g(x) \quad \text{for all } x \in X,$$

there is a least element (morphism), and any increasing ω-chain of morphisms has a least upper bound.

SP show that the property that $(C, \{e_i : X_i \to C\}_{i\in\omega})$ is an ω-colimit of an ω-chain is equivalent to a property about the least-upper bound of the maps $\{e_i\}_{i\in\omega}$. That is, the *global* property of being a colimit is equivalent to a *local* property about the hom-sets. As a consequence, we can show that a functor preserves ω-colimits by considering just how it acts on morphisms in the hom-sets, which can be substantially easier. These *locally monotonic* and *locally continuous* conditions are in Def. 9 and Def. 10 in SP.

1. Let $1_C : \mathbf{CPO}_\perp \to \mathbf{CPO}_\perp$ be the constant functor that maps every CPO to the fixed CPO C, and every morphism to the identity morphism id_C. Show that this functor is locally monotonic and locally continuous.

2. Let $+ : \mathbf{CPO}_\perp \times \mathbf{CPO}_\perp \to \mathbf{CPO}_\perp$ be the sum functor described in Example 2 of Section 5 in SP. Show that this functor is locally monotonic and locally continuous.

3. Let $\to_\perp : \mathbf{CPO}_\perp^{op} \times \mathbf{CPO}_\perp \to \mathbf{CPO}_\perp$ be the strict function-space functor described in Example 2 of Section 5 in SP. Show that this functor is locally monotonic and locally continuous.

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1. The functor $1_C(X) = C$, $1_C(f) = \mathrm{id}_C$. The map $\mathbf{CPO}_\perp(Y, Z) \to \mathbf{CPO}_\perp(1_C Y, 1_C Z) = \mathbf{CPO}_\perp(C, C)$ sends every morphism to id_C, a constant map. Constant maps are monotone (preserving $\leq$) and continuous (joins of constants are constant). Thus, locally monotone and continuous.

2. The sum $X + Y$ is the disjoint union with bottoms identified, ordered componentwise, bottom the shared $\perp$.

   The functor $+ : \mathbf{CPO}_\perp \times \mathbf{CPO}_\perp \to \mathbf{CPO}_\perp$. For local properties, consider the action on homs. The map $\mathbf{CPO}_\perp(A, A') \times \mathbf{CPO}_\perp(B, B') \to \mathbf{CPO}_\perp(A{+}B, A'{+}B')$ sends $(f, g) \mapsto f{+}g$, where $(f{+}g)(\mathrm{inl}\ x) = \mathrm{inl}\ f(x)$, $(f{+}g)(\mathrm{inr}\ y) = \mathrm{inr}\ g(y)$, and $(f{+}g)(\perp) = \perp$. This is monotone: if $f \leq f'$, $g \leq g'$, then $f + g \leq f' + g'$ pointwise.

   For continuous: directed joins in product homs are pointwise, $(\bigsqcup f_i) + (\bigsqcup g_i) = \bigsqcup(f_i + g_i)$, since joins in sum are componentwise.

   Thus, locally monotone and continuous.

3. $[X \to_\perp Y]$ is the CPO of strict continuous functions $X \to Y$, ordered pointwise. The bifunctor $\to_\perp$: $\mathbf{CPO}_\perp^{\mathrm{op}} \times \mathbf{CPO}_\perp \to \mathbf{CPO}_\perp$. morphism from (X, Y) to (X', Y') is $(p : X' \to X, g : Y \to Y')$, and acts as $\phi \mapsto g \circ \phi \circ p$. The hom order in op is reversed in first component. The map is monotone: suppose $(p, g) \leq (p', g')$ in op $\times$ cov, i.e., $p' \leq p$, $g \leq g'$. Then for ϕ, $g \circ \phi \circ p \leq g' \circ \phi \circ p'$? Since $p' \leq p$, $\phi \circ p' \leq \phi \circ p$ (monotone), then $g \circ (\phi \circ p') \leq g \circ (\phi \circ p) \leq g' \circ (\phi \circ p)$ (if g monotone), but need adjustment.

Since composition is monotone in both arguments (pre and post), and reversal in op, it preserves the order. This is standard in domain theory.

For continuous: joins are preserved because composition preserves joins (continuous maps).

Thus, locally monotone and continuous.

Exercise 3. A pair of morphisms $e : X \to Y$ and $p : Y \to X$ is an *embedding-projection* pair if $p \circ e = id_X$, and $e \circ p \sqsubseteq id_Y$. Theorem 3 in SP (Theorem 4.13 in FPP) shows that we have ω-colimit preserving functors:

- $1_C : \mathbf{CPO}^E_\perp \to \mathbf{CPO}^E_\perp$
- $+ : \mathbf{CPO}^E_\perp \times \mathbf{CPO}^E_\perp \to \mathbf{CPO}^E_\perp$
- $\to_\perp : \mathbf{CPO}^E_\perp \times \mathbf{CPO}^E_\perp \to \mathbf{CPO}^E_\perp$ [*not* a typo: covariant]

where $\mathbf{CPO}^E_\perp$ is the subcategory with the same objects, but only *embeddings* for morphisms. You can assume that $\mathbf{CPO}^E_\perp$ has all ω-colimits.

Show that the following equation has a solution in $\mathbf{CPO}^E_\perp$:

$$D \cong \mathbb{B}_\perp \cup (D \to D)$$

where $\mathbb{B}_\perp$ is the CPO with elements $\{tt, ff, \perp\}$ where $\perp \sqsubseteq tt$ and $\perp \sqsubseteq ff$ (the *flat* or *lifted* booleans); $\cup$ is sum in $\mathbf{CPO}^E_\perp$, and $\to$ is the strict function-space in $\mathbf{CPO}^E_\perp$.

Since $\mathbf{CPO}^E_\perp$ is a subcategory of $\mathbf{CPO}_\perp$ with the same collection of objects (a *wide* subcategory), the solution in $\mathbf{CPO}^E_\perp$ is also a solution in $\mathbf{CPO}_\perp$.

[Hint: using the pieces above, define functor $F : \mathbf{CPO}^E_\perp \to \mathbf{CPO}^E_\perp$ and show that it preserves ω-colimits (you can use any of the preservation properties at the end of Métayer's notes). Build a chain in $\mathbf{CPO}^E_\perp$, noting that the unique map $i : 0 \to F(0)$ is an embedding and F preserves embeddings, and take the colimit.]

Food for thought (not for credit). In $\mathbf{CPO}_\perp$, the initial object $\{\perp\}$ is also the terminal object (a category with this property is said to have a *zero object*. In fact any *strict* $\mathbf{CPO}_\perp$-enriched category has a zero object (see Lemma 4.4 in FPP). If an object C is the ω-colimit of the above chain, we can actually show that it is also the ω-*limit* of the following, reversed chain:

$$1 \xleftarrow{!} F(1) \xleftarrow{F!} F(F(1)) \xleftarrow{F(F!)} \cdots$$

where $(i, !)$ is an embedding-projection pair, and the maps $\{p_i : C \to X_i\}_{i \in \omega}$ are the projection pairs of $\{e_i : X_i \to C\}_{i \in \omega}$. We showed in class that colimits give initial algebras and least fixed-point of F; likewise, the limit of this chain gives the final co-algebra and greatest fixed-point of F. Since the object C is both a

colimit of the initial algebra chain and also a limit of the final coalgebra chain, the fixed point of F is unique up to isomorphism in $\mathbf{CPO}_\perp$. This is usually not the case in other categories, where the least and greatest fixed-points are usually not isomorphic.

Finally, a similar strategy can be used to show the existence of solutions to equations in the category of metric spaces. See the paper "Solving reflexive domain equations in a category of complete metric spaces" by America and Rutten (JCSS, 1982), or Section 5 of FPP.

The functors $1_C, +, \to_\perp$ preserve ω-colimits in $\mathbf{CPO}^E_\perp$. Define $F : \mathbf{CPO}^E_\perp \to \mathbf{CPO}^E_\perp$ by $F(X) = B_\perp + (X \to_\perp X)$, a composition of the given functors (constant $B_\perp$, sum, function space). Since each preserves embeddings (locally monotone implies preserves embeddings) and ω-colimits, F does too. Let 0 be the initial object $\{\perp\}$. The unique $i : 0 \to F(0)$ is an embedding since $\mathbf{CPO}^E_\perp$ has ω-colimits. Build the chain: $0 \xrightarrow{i} F(0) \xrightarrow{F(i)} F^2(0) \xrightarrow{F^2(i)} \cdots$. The colimit $D = \operatorname{colim} F^n(0)$ exists. Since F preserves ω-colimits, $F(D) \cong D$. Thus, a solution in $\mathbf{CPO}^E_\perp$, hence in $\mathbf{CPO}_\perp$.

is an embedding because colimits exist? not sure.

Exercise 4 (Category of the Week). In this exercise, we'll give a quick introduction to *enriched categories*. This is a rich branch of category theory that we cannot hope to cover here, but we will aim to give the main definitions and work out a few examples of enriched categories. Max Kelly's book, Basic Concepts of Enriched Category Theory, is a classic reference (although it is not really "basic").

In a nutshell, enriched category theory generalizes the set (or collection) of morphisms $\mathbf{C}(A, B)$ in standard category theory to be an object in some category V, called the enriching category. In this way, the morphisms from A to B can themselves have some structure (e.g., they might have an order), and composition of morphisms is required to respect this structure.

The most basic notion, described here, is a *category enriched in a monoidal category* $\mathbf{V}$, also known as a $\mathbf{V}$-category. You can think of $\mathbf{V}$ as a standard category. We will see the definition of a monoidal category in the upcoming weeks, but for now, you can think of it as a category where for any two objects A and B, there is another object $A \otimes B$. Importantly, this object is *not* assumed to be the Cartesian product—a monoidal product isn't defined via a universal property, and so a given category might have multiple, non-isomorphic monoidal products, and we do not assume/require that there are projections from the object $A \otimes B$.

1. Given two CPOs $(X, \sqsubseteq_X)$ and $(Y, \sqsubseteq_Y)$, the *smash product* is the CPO with underlying set given by the product of underlying sets of X and Y and pointwise order, except with the $(\perp_X, y) = (x, \perp_Y) = \perp$, i.e., a pair with either element equal to bottom is treated as the bottom element. It is known that the smash product is a monoidal product on $\mathbf{CPO}_\perp$.

Starting from the order on morphisms in Exercise 2, show that the (standard) category $\mathbf{CPO}_\perp$ can be enriched in $\mathbf{CPO}_\perp$ (viewed as a monoidal category with the smash product).

2. For a different sort of example, let $\mathbf{R}_+$ be the category with set of objects $r \in [0, \infty]$, and there is a morphism $f : r \to s$ if and only if $r \geq s$; since there is at most one morphism between any two objects, $\mathbf{R}_+$ is a posetal category. Let $r \otimes s$ be defined to be $r + s$; this operation makes $\mathbf{R}_+$ into a monoidal category.

   Show that any metric space (M, d) can be encoded as a $\mathbf{R}_+$-enriched category, also known as a *Lawvere metric space*. In fact, a $\mathbf{R}_+$-enriched category generalizes some of the typical metric space axioms: which of the usual metric space axioms are relaxed in a Lawvere metric space?

3. Enriched category theory is a generalization of standard category theory: along with enriched categories, there are also enriched versions of functors, natural transformations, limits/colimits, etc. For just a glimpse, suppose $\mathbf{C}$ and $\mathbf{D}$ are $\mathbf{V}$-categories. Take a look at the definition of an enriched functor from $\mathbf{C}$ to $\mathbf{D}$. Unfolding these definitions, show that $\mathbf{R}_+$-functors are non-expansive functions between (Lawvere) metric spaces.

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1. The smash product $X \otimes Y$ has underlying set $X \times Y$, pointwise order, but $(\perp_X, y) = (x, \perp_Y) = \perp$ for the new bottom. The hom-object $\mathbf{CPO}_\perp(X, Y)$ is the CPO of strict continuous functions, pointwise order, bottom the constant $\perp_Y$. Composition: $\mathbf{CPO}_\perp(Y, Z) \otimes \mathbf{CPO}_\perp(X, Y) \to \mathbf{CPO}_\perp(X, Z)$, $(f, g) \mapsto f \circ g$ (if one is bottom, output bottom, which matches constant $\perp \circ g = \perp$). Identity: unit I (two-element CPO $\perp < \top$) maps to id_X. This respects the enrichment, as composition is continuous and monotone. Thus, enriched.

2. Consider the category $\mathbf{R}_+$ with objects the extended non-negative reals $[0, \infty]$, a unique morphism $r \to s$ iff $r \geq s$, and tensor $r \otimes s = r + s$ (monoidal unit 0). A $\mathbf{R}_+$-enriched category consists of: A set M of objects (points), For each pair $m, m' \in M$, a hom-object $d(m, m') \in [0, \infty]$, Composition: for m, m', m'', $d(m', m'') + d(m, m') \geq d(m, m'')$ (triangle inequality), Identities: $0 \geq d(m, m)$ (so $d(m, m) = 0$). This encodes a metric space (M, d), where d is the hom.

   For metric (M, d): hom-object $d(x, y) \in \mathbf{R}_+$, composition $d(y, z) + d(x, y) \geq d(x, z)$ (triangle), id $d(x, x) = 0$. Relaxed axioms: No symmetry: $d(m, m')$ not need equal $d(m', m)$; No separation: d(m,m')=0 need not imply $m = m'$; Allows $d = \infty$.

3. An enriched functor $F : \mathcal{C} \to \mathcal{D}$ (V-enriched) assigns objects, and V-morphisms $\mathcal{C}(A, B) \to \mathcal{D}(FA, FB)$, preserving composition and ids. For $\mathbf{R}_+$-enriched (Lawvere metrics), the hom-map $d_\mathcal{C}(a, b) \to d_\mathcal{D}(Fa, Fb)$ is a

morphism in $\mathbf{R}_+$, i.e., $d_{\mathcal{D}}(Fa, Fb) \leq d_{\mathcal{C}}(a, b)$ (since $r \to s$ iff $r \geq s$). Thus, $d(Fa, Fb) \leq d(a, b)$, non-expansive.
